

Hackathon Project Brief: City Dispatch Workflow & Fleet Automation

1. Challenge Overview & Industry Context

Small-to-medium regional carriers operate under heavy software fragmentation. A typical dispatcher for a **City Dispatch Platform** must manage short-haul and regional runs while switching between **~5 disjointed software systems** every single day:

- **Load / Contract Boards:** ~\$800/month per device
- **Electronic Logging Devices (ELD) / Logbooks:** ~\$600/year per truck
- **Transportation Management System (TMS / Dispatch):** ~\$50/month per user
- **Quoting & Rate Tools:** Additional subscription fees
- **Maintenance & Accounting Systems:** Separate enterprise setups

This constant context-switching creates massive operational bottlenecks. Dispatchers spend hours manually micro-managing drivers, verifying availability, and making "check calls" instead of focusing on high-value quoting and spot market revenue.

Freight Terminology Primer:

- **FTL (Full Truck Load):** A shipment that occupies an entire semi-trailer or dedicated truck. FTL orders go directly from origin to destination without intermediate stops or handling.
- **LTL (Less-Than-Truckload):** Smaller freight shipments that do not fill an entire trailer. Multiple shippers' goods are consolidated onto one truck, requiring multi-stop routing and terminal transfers.

The Financial Impact:

Each missed load quote is worth **\$1,000 to \$7,000**. When dispatchers are bogged down by manual workflows, a fleet misses roughly **1 to 3 jobs per day**. That equals roughly **\$6,000 per day** in missed opportunities—accumulating to an estimated lost revenue of **\$15,000 to \$50,000+ monthly** for a single dispatch desk!

The Hackathon Goal:

Historically, carriers resisted unified dashboards due to high custom development costs and lengthy implementation timelines. **Your goal is to prove to industry leaders that fully integrated, intelligent dispatch dashboards can be built rapidly and affordably in the AI revolution.**

2. Operating Region: Southern Ontario City Dispatch

Participants should model their systems around a **City Dispatch / Regional Freight Platform** serving the Southern Ontario logistics corridor.

- **Primary Terminal Hubs:** London, ON and Milton, ON
- **Coverage Boundaries:**
 - **North:** Barrie, ON
 - **East:** Peterborough, ON & Pickering, ON
 - **West:** London, ON
 - **South:** Niagara Falls, ON

Dispatchers in this region constantly route high-density FTL and LTL regional runs between major distribution hubs along the 400-series highways (e.g., Highway 401, 403, and 400).

3. Target Architecture & Key Features

System Component 1: Web Dispatcher Dashboard with Mapping API

- **Production Mapping & Satellite Toggle:** Integrated Google Maps, Mapbox, or Leaflet/OpenStreetMap engine with a **Satellite View toggle** so dispatchers can visually inspect dock layouts, driveway entrances, yard space, and physical obstacles in industrial parks across Southern Ontario.
- **Track & Trace Suite:**
 - **Historical Breadcrumbs:** Visualizes exact historical routes taken across the regional grid.
 - **Distance & Odometer Tracking:** Tracks total miles/kilometers driven per leg and shift.
 - **Speed Telematics:** Records live and historical driver speeds to identify traffic bottlenecks or delay events.
- **Geofences & Detention Automated Billing (Critical Requirement):**
 - **The Problem:** In FTL freight, carriers typically offer the **first 2 hours of loading/unloading dock time for free**. Any time spent waiting beyond 2 hours is billed as **detention time**. If dispatchers miss logging arrival or completion times, carriers lose thousands in unbilled detention fees.

- **The Database Solution:** Whenever a truck crosses a geofence around a facility, the system must automatically record an **exact timestamp in the database** for when the driver **arrived** at the dock and when they **finished/departed**. The dashboard uses these timestamps to auto-calculate detention fees owed after the 2-hour threshold.

System Component 2: The Separated Simulation Engine

- **Fleet Telematics Simulator:** Backend service that simulates live truck movements along Southern Ontario road networks, updating GPS coordinates, current speed, odometer readings, and HOS log states in real time.
- **Event Generator:** Simulates route delays (e.g., 401 highway slowdowns), dock wait times, and duty cycle shifts to stream live tracking data directly to the portal.

System Component 3: Driver Interface (Mobile App or Responsive Web)

- **Digital Logbooks (ELD) & Dispatch View:** Tracks driver duty status, receives incoming load specs, and provides route maps.

4. Canadian Hours of Service (HOS) & Logbook Primer

Commercial vehicle drivers operating in Southern Ontario must log their Record of Duty Status (RODS) according to **Transport Canada Regulations (South of 60°N)**:

- **13-Hour Driving Limit:** Drivers can drive a maximum of **13 hours** within a work shift.
- **14-Hour On-Duty Limit:** Drivers cannot drive after accumulating **14 hours of total on-duty time** (driving + non-driving work like loading, unloading, or waiting at docks).
- **16-Hour Elapsed Window:** No driving is allowed after **16 hours have elapsed** since the end of the last 8+ consecutive hour rest break.
- **10 Hours Daily Off-Duty:** Drivers must have at least **10 hours off-duty daily**, including an 8-consecutive-hour core rest period.
- **Cycle Limits:** Carriers operate under **Cycle 1** (70 hours on-duty in 7 days) or **Cycle 2** (120 hours on-duty in 14 days).

Data Resources: Transport Canada Commercial Vehicle Drivers Hours of Service Regulations, MTO Commercial Driver's Handbook, and CCMTA ELD standards.

5. Key Problem Areas & "Bug/Edge Case" Discovery

Teams are encouraged to actively hunt for operational "bugs" in city dispatch workflows:

- **Automated Detention Billing:** Timestamping dock arrivals and departures in the database to automatically bill shippers for detention time beyond the free 2-hour FTL allowance.
- **Deadhead & Empty Mile Reduction:** Proximity-based load matching that automatically queues a return LTL or FTL load (e.g., pairing a Milton-to-London delivery with a London-to-Kitchener return shipment).
- **Automated HOS & Axle-Weight Compliance:** Pre-dispatch audits cross-referencing driver logbook availability (13h/14h limits) and truck weight limits before assigning a route.
- **Edge Case / Hidden Problem Discovery (Bonus Focus):** What happens when a driver runs out of HOS while sitting in a dock queue waiting for a load? What if a truck encounters unexpected 401 highway closures? Finding and solving these real-world edge cases elevates your project score.

6. Technical & Functional Deliverables

Component	Required Core Functionality
Dispatcher Dashboard	• Interactive Southern Ontario map with Satellite View toggle • Track & Trace Suite: Route breadcrumbs, total distance driven, speed logs • Geofence & Database Logging: Timestamped arrival/departure database records for automated detention billing (>2 hrs) • Automated load-matching and HOS compliance engine

Component	Required Core Functionality
Simulation Engine	• Independent script/service simulating live truck movement along Southern Ontario routes • Telemetry updates (coordinates, speed, distance, HOS elapsed)
Driver Interface	• Mobile app or responsive web view for load acceptance and duty log tracking
Pipeline & Integration	• Multi-system consolidation (combining load board, ELD, TMS, and detention tracking)

7. Judging Criteria

Projects will be evaluated based on the following qualitative areas:

- **Workflow Speed & Financial Value:** How effectively does the solution eliminate manual dispatch overhead so dispatchers can focus on quoting high-revenue jobs (\$15k–\$50k/month recovery potential)?
- **Geofencing & Detention Precision:** Correct database recording of arrival/completion timestamps to calculate dock wait times and automate post-2-hour detention billing for FTL runs.
- **Problem Discovery & Innovation:** How deeply did your team explore the problem space? Did you uncover subtle edge cases, hidden industry pain points, or "software bugs" in standard city dispatch operations?
- **Mapping & Track & Trace Depth:** Quality of real-world mapping across the Southern Ontario regional boundary, satellite mode, route breadcrumbs, distance tracking, and speed telemetry.
- **HOS & Regulatory Logic:** Practical execution of logbook checks aligned with Canadian 13h/14h regulations.

- **Simulation Engine & Real-Time Sync:** Realism of the dedicated simulator in modeling truck movements, speed variations, dock wait times, and geofence triggers.
- **UI/UX & Usability:** Is the consolidated dashboard intuitive, clean, and capable of replacing fragmented legacy tools?